

The Problem of Digital Dualism in Social Virtual Reality Research: Toward a Hybrid Culture, Space, and Body

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ABSTRACT

The problem of digital dualism represents a bias that conceptualizes physical and virtual realities as separate entities, thereby influencing the methodological approaches employed in research on these domains. Digital dualism has been evident since the earliest studies of cyberspace. This position paper presents how this bias is inadvertently reproduced in research on social virtual reality (VR), focusing on issues such as the omission of the influence of physical environments, the neglect of users' geographical, ethnic, and socio-cultural situatedness, the disproportionate reliance on laboratory studies over field research, and in the conceptualization of the user-avatar relationship. Addressing these research gaps, rooted in digital dualism, is essential for advancing future investigations in social VR.

Index terms: digital dualism, virtual reality, hybrid, culture, space, body

1 INTRODUCTION

The question of the relation between physical and virtual reality (VR) constitutes a philosophical problem. Nevertheless, when designing any research concerning VR, we inevitably – whether intentionally or not – adopt a heuristic response to this question, a stance that shapes both the research design and its subsequent outcomes. The purpose of this position paper is to broaden the discussion on the reciprocal influence of physical and virtual environments by highlighting the broader issue of digital dualism. This issue pertains to the methodological assumptions in VR studies, which (implicitly or explicitly) frame virtual and physical environments as distinct from one another. On the one hand, such assumptions lead to the omission of significant aspects in research. On the other hand, they may create a self-reinforcing cycle in some cases, as these assumptions hinder the collection of data that could undermine their validity.

Digital dualism, a concept popularized by Jurgenson, refers to a bias that, firstly, treats online and offline spaces as separate, and secondly, privileges life within physical reality as ‘real’ while regarding life within digital reality as ‘less real’ [1], [2]. In this paper, I will focus on the first aspect of this bias. While digital dualism is primarily rooted in cultural imaginaries of cyberspace, its influence on research design can affect both the data collected and the conclusions drawn.

From the inception of Internet studies, many scholars characterized cyberspace as a disembodied domain, and thus inherently liberated from the constraints of corporeality and the oppressions associated with embodied identities, such as gender,

age, race, and disability [3], [4]. Other scholars conceptualized cyberspace as a post-national domain where geographical and cultural differences would diminish in significance [5], [6]. These conceptualizations of cyberspace were rooted in its perceived separation from the physical world, physical bodies, and physical territories. Jurgenson further highlights that dystopian visions of cyberspace – characterized by the perception of interactions within it as illusory or deceptive – also arise from the same foundational framing of cyberspace as distinct from physical reality [1].

This position paper identifies four research gaps in the study of social VR that stem from the perpetuation of digital dualism in existing research. Implementing changes in methodological approaches within these areas is essential for advancing future research in social VR. While, due to paper constraints, this article focuses exclusively on examples from social VR studies, it highlights a broader issue – the problem of digital dualism in VR research.

2 DIGITAL DUALISM IN SOCIAL VR RESEARCH

Since the inception of research on social VR, digital dualism has been reproduced within this domain, resulting in underexplored reciprocal influences between virtual and physical environments. At the outset, it is crucial to emphasize that the distinction between VR and mixed reality (MR) is, in itself, problematic. MR environments are commonly defined as those “where the real and the virtual coexist” [7], in contrast to VR environments, which are described as entirely virtual. While the difference between the two is clear, this division unintentionally produces the assumption that a VR user exists solely within a computer-generated world, as opposed to an MR user. Acknowledging the seemingly self-evident fact that a VR user, even while wearing a head-mounted display, is always situated within a hybrid or mixed physical-virtual environment constitutes a critical first step toward overcoming digital dualism in VR research.

2.1. Omission of Physical Spaces in Social VR Research

To the best of the author’s knowledge, there are no field studies on social VR that specifically aim to investigate the impact of users’ physical environments on their interactions within social VR. This constitutes a significant research gap, highlighting the presence of digital dualism in past studies, that frame social VR interactions as separate from users’ physical environments.

The literature provides isolated examples suggesting that addressing this gap might be essential for deepening our understanding of these interactions. For instance, Freeman et al. [8] report that several individuals in their study behaved differently in social VR depending on whether other household members were present in their home or whether they were alone in their physical space. The use of VR technology makes users’ behaviors more visible in physical spaces for others compared to desktop technologies. Although this was not the focus of Freeman et al.’s study [8], it highlights a significant underexplored area: the

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influence of the presence of other household members on interactions in social VR.

The synchronization of the movements of the digital body with the physical body ensures that user interactions in social VR are not only visible in both spaces but also actively engage with both spaces simultaneously. This is particularly evident in activities such as physical fitness or dance, which might require users to adapt their interactions to the limitations of physical spaces or to modify the physical spaces themselves to accommodate these interactions [9], [10], [11]. Saker and Frith [12] highlight the need to reconceptualize the relationship between physical and digital spaces in the VR research, proposing that it be understood as coextensive rather than dualistic. This area of research is also significant given the trend of decreasing apartment sizes in some parts of the world, which may limit the use of VR technology in the future or necessitate the development of new strategies by users to address these constraints [13]. In the future, it seems important to develop research that places greater emphasis on exploring the mutual influence between physical and digital environments, thereby transcending digital dualism.

2.2. The Avatar – User Relationship

The reconceptualization of the relationship between the user and avatar is one of the key challenges for future research, particularly with the emergence of new phenomena in social VR, such as phantom bodily sensations and embodied forms of intimacy and aggression [11], [14], [15], [16], which transcend the dualistic distinction between users' physical body and avatar.

The concept of the avatar, understood as a virtual representation of the user, inherently presupposes an a priori distinction between two entities: the physical user and their virtual representation [11], [17]. However, user experiences and behaviors observed in empirical studies seem to challenge this dualistic conceptualization. Firstly, contemporary research on phenomena such as the Proteus Effect and the body semantic effect demonstrates that the avatar not only represents the user but also actively influences who they are in VR – shaping their behavior, attitudes, and cognitive functions [18], [19], [20], [21]. Secondly, Krell and Wettmann argue that in many practices within social VR, users experience their digital bodies as extensions of their physical embodiment rather than as separate entities [14].

There is a need for the further development of theoretical frameworks capable of capturing the hybrid nature of users' embodied experiences in social VR, transcending digital dualism. In another study, I propose conceptualizing the relationship between the physical body, the digital body, and technology in VR as cyborgic and context-dependent [11]. The idea of using the concept of the cyborg to describe the hybrid embodiment of the user in VR within computer-mediated communication is not new [22]; however, it has received relatively little attention in the academic literature to date.

2.3. The Omission of Physical Ethnicity

Another issue that may stem from digital dualism is the matter of the limited balance between users from different parts of the world in field research. This issue is tied to the fact that the vast majority of field research in social VR has been conducted in English. As a result, the perspectives of users from English-speaking countries may be privileged in such studies, while those from non-English-speaking regions might be marginalized.

The experiences of these two user populations may differ, as I have noted in another paper [23]. My ethnographic research underscores, using the case of Polish users, the emerging communication challenges encountered by non-native English speakers in social VR, which have not been previously documented in the literature. This study also highlights the

tendency of national and linguistic diasporas to form within social VR among non-native English speakers, driven by barriers related to the limited accessibility of global spaces for these users. These diasporas often become arenas for the (re)production of ethnic identities; however, due to the predominant focus of research on English-speaking communities, these topics have received little scholarly attention.

Furthermore, Zhang [24] highlights differences in the socio-cultural practices of Asian and Caucasian users within social VR. These examples demonstrate that even though users create digital cultures within social VR, this process remains influenced by their offline cultures. In digital ethnography, researchers emphasize the need for situated exploration of how users' cultural differences shape their engagement with digital technologies [25]. However, the impact of users' offline cultural situatedness on their interactions within social VR remains largely underexplored.

MacArthur et al. emphasize the necessity of developing a critical approach in social VR research, citing observed “disparities between dominant and non-dominant groups concerning some intersecting facets of identity, such as race and gender” [26]. Ensuring balance between users from dominant and non-dominant groups is a crucial step for future studies in this domain. Additionally, it is essential to maintain balance among users from various ethnic groups and geographical regions to achieve a deeper understanding of interactions in social VR without marginalizing minority voices.

2.4. Research in Laboratory Settings

The majority of studies on communication in social VR are conducted in laboratory settings [27]. This may further exacerbate the previously discussed limitations of research rooted in the problem of digital dualism.

Firstly, VR studies conducted in laboratory spaces limit the ability to examine how users' natural physical environments influence their interactions within social VR platforms. Secondly, recruiting participants from diverse regions and ethnic backgrounds presents significant challenges in laboratory settings. This may result in a limitation in the ability to capture the influence of factors related to differences in users' offline cultural situatedness on interactions in social VR. Thirdly, conducting VR studies in laboratory settings makes it difficult to capture challenges related to geographical distance between users, such as internet connectivity issues or time zone differences, which are encountered in everyday usage [23].

Another issue, which appears to be rooted in digital dualism and becomes apparent in laboratory research, is the tendency to assess the effectiveness of new software exclusively within controlled environments. This approach inadvertently includes the assumption that virtual environments should be experienced uniformly, regardless of the physical context in which they are accessed. Despite the rapid development of social VR platforms in recent years, there remains a disproportionately small number of studies that evaluate the effectiveness of these tools in real-world environments [27]. Such environments present unique challenges, including spatial constraints, internet connectivity issues, and limitations related to time and personal resources. Addressing these limitations in future research is essential for the advancement of social VR technologies.

3 CONCLUSIONS

The identified areas demonstrate that digital dualism is still unintentionally reproduced in VR research, thereby contributing to the emergence of the discussed research gaps. Addressing these underexplored areas is crucial for future studies. Firstly, future field research on social VR should place greater emphasis on how

users' offline situatedness influences their interactions within virtual environments. Reconceptualizing the user-avatar relationship in VR is essential. Secondly, sample selection in studies must prioritize achieving a balance among participants from diverse geographical, ethnic, cultural, and social backgrounds. Thirdly, future research should increasingly be conducted in natural environments where VR technologies are intended for use. This approach would enable a better understanding of how the constraints of such environments – including spatial, technical, and resource limitations – affect the effectiveness of these tools. Addressing these aspects will enable future studies to provide a more nuanced and holistic understanding of social VR, thereby mitigating the limitations imposed by digital dualism.

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